

HAO TAN

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EDUCATION

Guangdong University of Technology (GDUT)

Sep 2022 – Jun 2026

Bachelor of Engineering in *Information Engineering*

GPA: 4.1/5.0 (91.03%), **Rank:** 3/190 (Top 2.1%)

Research Interests: *Efficient AI, Multimodal Generative Model, Multimodal Large Language Model, Deepfake Detection, World Model, AI for Science*

PERSONAL HONORS

[2025] Top Ten Outstanding Undergraduates of Guangdong University of Technology (10 university-wide)

[2025] National Scholarship

[2025] Shiyuan Corporate Scholarship, First Prize (*rk 1/2262*)

[2025] Outstanding Students Scholarship, First Prize (*rk 1/553*)

[2025] Top Academic Performance Scholarship

[2025] Innovation Scholarship

[2024] Outstanding Students Scholarship, Second Prize (*rk 11/553*)

[2024] Innovation Scholarship

[2023] Outstanding Students Scholarship, Second Prize (*rk 37/553*)

PUBLICATIONS

[1] PSCNet: Long Sequence Time-series Forecasting for Photovoltaic Power via Period Selection and Cross-variable Attention. *Applied Intelligence (APIN)*. (**SCI Q3, First Author**) [\[link\]](#)

[2] Adaptive Few-shot Prompting for Machine Translation with Pre-trained Language Models. *AAAI 25*. (**CCF-A, Fourth Author**) [\[link\]](#)

[3] HLF-CIKNet: A Spatial-Frequency and Cross-Interaction Feature Extraction Network for EEG-Based Emotion Recognition. *Biomedical Signal Processing and Control (BSPC)*. (**SCI Q2 Top, First Author, Under Review**)

[4] TDAF-Net: A Temporal Difference Alignment Fusion Network for Multimodal Deception Detection. *CVPR 26 The 2nd Workshop & Challenge on Subtle Visual Computing*. (**CCF-A Workshop, Fourth Author, Under Review**)

[5] Source-Free Domain Adaptation based on Physiology-specific Disentanglement for Motor Imagery EEG Classification. *Engineering Applications of Artificial Intelligence (EAAI)*. (**SCI Q1, First Author, Under Review**)

[6] One-shot Unsupervised Domain Adaptation for Semantic Segmentation with Self-Constrained Style Decoupling. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*. (**SCI Q1, First Author, Under Review**)

[7] MFEFN: Multimodal Feature Enhancement and Fusion Network for Multimodal Sentiment Analysis. *Pattern Recognition (PR)*. (**SCI Q1, Fourth Author, Under Review**)

[8] Multi-level Denoising and User Preference Alignment for Multimodal Recommendation. *Expert Systems With Applications (ESWA)*. (**SCI Q1, Third Author, Under Review**)

INTELLECTUAL PROPERTY

[2025] *CN202411788172.X*. Methods and Systems for Sports Injury Risk Detection and Risk Location Identification (Invention Patent Grant, Second Inventor)

[2025] *CN202411588223.4*. A Photovoltaic Power Generation Forecasting Method and System Based on Frequency Decoupling and Cross-Attention (Invention Patent Publication, Second Inventor)

[2025] *CN202510523099.1*. A Water Environment Monitoring Method Based on Multi-View Features and Knowledge Distillation (Invention Patent Publication, Second Inventor)

[2025] CN202411798679.3. A Cross-Subject EEG-Based Emotion Recognition Method and System Using Spatial Interaction Features (Invention Patent Publication, Second Inventor)

[2025] CN202411618399.X. An Intelligent Anti-Theft Food Delivery Storage Device and Its Usage Method (Invention Patent Publication, Third Inventor)

[2024] 2024SR1630302. Sentiment Analysis Software for Reviews Based on an Improved Transformer (Software Copyright Grant, Second Author)

[2024] 2024SR1218215. Takeaway Mat Software Based on MAML Meta-Learning (Software Copyright Grant, Third Author)

[2024] 2024SR0368163. Swarm UAV Cooperative Task Planning Software Based on Particle Swarm Optimization (Software Copyright Grant, First Author)

[2024] 2024SR0367978. Path Planning Software for Complex Mountain Roads Based on the Astar Algorithm (Software Copyright Grant, First Author)

[2024] 2024SR0721670. Earthquake Magnitude Prediction Software Based on the Holt-Winters Algorithm (Software Copyright Grant, Second Author)

[2024] 2024SR0752622. Base Station Layout Optimization Software Based on the Spider Wasp Optimization Algorithm (Software Copyright Grant, Second Author)

[2024] 2024SR0108668. Face Feature Recognition and Analysis Software Based on Computer Vision (Software Copyright Grant, First Author)

RESEARCH PROJECTS

[2025] An Intelligent Early Warning System for Sports Injuries Based on Multi-Source Biosignals (National Undergraduate Innovation Training Program, Project Leader)

[2025] Water Environment Monitoring System Based on Multi-View Features and Knowledge Distillation (National Undergraduate Innovation Training Program, Participant)

[2024] A Multimodal Recommendation System Guided by Disentangled Modality Co-Occurrence Signals (National Undergraduate Innovation Training Program, Participant)

[2024] An Intelligent Visual Reconstruction System Based on Hierarchical Encoding of EEG Signals (National Undergraduate Innovation Training Program, Participant)

[2023] A Cultural Domain-Specific Dialogue Model Based on Large Language Models (Provincial Undergraduate Innovation Training Program, Participant)

[2023] Intelligent Wiring Identification and Verification System for Electrical Cabinets Based on OCR Technology (University-Level Undergraduate Innovation Training Program, Participant)

AWARDS

[2025] Kaggle 2025 Drawing with LLMs Challenge, Silver Medal (*Global rk 48/1313*)

[2025] The 7th Global Campus AI Algorithm Innovation Competition, National First Prize

[2025] The 27th China Robotics and AI Algorithm Innovation Competition, National Third Prize

[2025] The 7th Global Campus AI Algorithm Innovation Competition Guangdong Region, Second Prize

[2025] The Milan Design Week – China Higher Education Design Exhibition, National Third Prize

[2025] AIC 2025 Webly Supervised Fine-Grained Image Recognition Challenge Guangdong Region, First Prize

[2025] AIC 2025 Smart Meteorology Track National Selection Round, Third Prize

[2025] China Creative National Digital Art and Design Competition Guangdong Region, First Prize

[2025] The 18th National College Computer Design Competition Guangdong Region, Second Prize

[2025] The 16th Lanqiao Cup Python Programming Track Guangdong Region, First Prize

[2025] The 27th China Robotics and AI Algorithm Innovation Competition Guangdong Region, Second Prize

[2025] The 27th China Robotics and AI Algorithm Innovation Competition Guangdong Region, Third Prize

[2025] The Milan Design Week – China Higher Education Design Exhibition Guangdong Region, Third Prize

[2024] Mathematical Contest in Modeling (MCM), Meritorious Winner

[2024] The 2nd Greater Bay Area AI for Science Competition, National Second Prize

[2024] The 6th Global Campus AI Algorithm Innovation Competition, National Third Prize

[2024] AIC 2024 Breast Ultrasound BIRADS Classification Challenge, National Third Prize

[2024] The 6th Global Campus AI Algorithm Innovation Competition Guangdong Region, First Prize

[2024] National College Mathematical Modeling Contest Guangdong Region, First Prize

[2024] RAICOM Robotic Developer Algorithm Optimization Challenge Guangdong Region, Third Prize

[2024] The 5th MathorCup Challenge on Mathematical Applications – Big Data Track, National Third Prize

[2024] The 2nd International College Mathematical Modeling Contest, Finalist

[2024] The 4th Tianfu Cup National College Mathematical Modeling Contest, National First Prize

[2024] The 5th Huashu Cup National College Mathematical Modeling Contest, National First Prize

[2024] The 10th National Statistical Modeling Competition Guangdong Region, Second Prize

[2024] The 3rd China College Big Data Challenge, National Third Prize

[2024] The 6th Zhongqing Cup National College Mathematical Modeling Contest, National Second Prize

[2024] The 10th Shuwei Cup International College Mathematical Modeling Challenge, Honorable Mention

[2024] The 16th Huazhong Cup College Mathematical Modeling Challenge, National Second Prize

[2024] The 3rd National Youth Innovation Translation Contest, National First Prize

[2024] The 15th Lanqiao Cup Embedded Design and Development Track Guangdong Region, Third Prize

[2024] The 15th Lanqiao Cup Digital Technology Innovation Track National Selection Round, Second Prize

[2024] The 11th Datang Cup ICT Engineering Practice Track Guangdong Region, Third Prize

[2024] The 9th Guangdong University of Technology Challenge Cup, Bronze Prize

[2024] The 5th GDUT Mathematical Modeling Competition, First Prize

[2024] The 9th China International College Students Innovation Competition GDUT School of Foreign Languages Selection Round, Third Prize

[2023] Huawei Ascend AI Innovation Competition Guangdong Region, Silver Prize

[2023] National College Mathematical Modeling Contest Guangdong Region, Second Prize

[2023] The 14th iCAN Innovation and Entrepreneurship Competition Guangdong Region, Third Prize

[2023] The 2nd College AI Technology Competition Innovation Application Track, National First Prize

[2023] The 13th Asia and Pacific Mathematical Contest in Modeling (APMCM), Second Prize

[2023] Shenzhen Cup Mathematical Modeling Challenge, Third Prize

[2023] The 4th GDUT Mathematical Modeling Competition, First Prize

[2023] The 12th Challenge Cup GDUT School of Foreign Languages Selection Round, Second Prize

CODE & CREATIONS

[2026] GenCard: a card generation tool based on the WanX model [\[github link\]](#)

[2026] Huaxia Basketball Manager: an AI-driven basketball simulation game set [\[github link\]](#)

[2026] DreamGravity: an AI-powered dream journaling and analysis tool [\[github link\]](#)

[2026] Imprint: an AI travel planning and diary system [\[github link\]](#)